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Mathematical statistics algorithm in the bending performance test of corroded reinforced concrete beams under fatigue load

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Abstract

Harsh environments such as alternating wet and dry conditions and cyclic loading cause erosion in steel bars, leading to severe damage to structures. Steel bars are also prone to bending under long-term fatigue loads. This paper establishes the bending performance test of corroded reinforced concrete (RC) beam under fatigue load based on the mathematical–statistical algorithm. It investigates the influence of heavy load on the fatigue performance of damaged RC beams, the influence of corrosive environment on the fatigue performance of RC beams, the degree of corrosion of steel bars in concrete beams under fatigue loading, and the distribution of chloride ions. The study results found that the stress ratio has a significant effect on the maximum crack width of the specimen beam. The greater is the stress ratio, the longer the fatigue life. This directly affects the performance of the specimen beam under fatigue loading. For this reason, we should pay special attention to the impact of corrosion on the fatigue performance of a structure during the design of steel bars.

Keywords: reinforced concrete beam, mathematical, statistical algorithm, fatigue load, bending performance**AMS 2010 codes:** 03B48

1 Introduction

Reinforced concrete (RC) beams are essential components of civil engineering structures such as buildings and bridges. Affected by corrosive environmental effects such as concrete carbonisation and chloride ion erosion, RC beams often suffer from steel corrosion and concrete expansion and cracking, which reduce their shear capacity. This affects the safety of the structure [1]. Therefore, the study and establishment of a calculation model for the shear capacity of corroded RC beams are of great significance for the safety assessment and bearing capacity redesign of in-service RC structures.

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The shear mechanism of corroded RC beams is complex and has many influencing factors. At present, research done on it has a few shortcomings: first, the introduced steel corrosion correction coefficient is usually determined according to engineering experience or test data fitting analysis. Second, the research model usually takes the critical oblique crack inclination angle to be approximately 45° , ignoring the impacts of factors such as shear span ratio, augmentation ratio, counterbearing ratio, and steel corrosion on the critical oblique crack inclination angle. This leads to limited calculation accuracy [2]. Therefore, it is necessary to comprehensively consider the influence of steel corrosion on the yield augmentation of counterbearings, counterbearing ratio, augmentation ratio, effective shear cross-sectional area of RC beams, and other essential factors. At the same time, we also need to consider the impact of objective uncertainty and subjective uncertainty to establish a probability model for calculating the shear capacity of corroded RC beams.

This paper first comprehensively considers the influence of steel corrosion on the yield augmentation of counterbearings, augmentation ratio, counterbearing ratio, critical oblique crack inclination, beam effective shear cross-sectional area, and other essential factors. At the same time, we established a deterministic model for calculating the shear capacity of corroded RC beams by combining the modified pressure field theory (MCFT) and the critical oblique crack inclination model, considering the influence of the shear span ratio [3].

2 Deterministic model for calculating the shear capacity of corroded RC beams

Corrosion of steel bars reduces the yield augmentation and effective cross-sectional area of counterbearings, leading to a decrease in the contribution of the counterbearing shear capacity and often causes concrete corrosion, expansion, cracking or cracking spalling [4]. The following comprehensive consideration of the influence of steel corrosion on the yield augmentation of counterbearings, augmentation ratio, counterbearing ratio, effective shear cross-sectional area of beams and other important factors will help to establish a deterministic model. Based on the MCFT, the shear capacity V of RC beams is mainly composed of the contribution value V_c of the concrete shear capacity and the contribution value V_s of the shear capacity of the counterbearings as:

$$V = V_c + V_s \quad (1)$$

After the steel is corroded, the cross-section will be rusted, which will cause stress concentration in the weakest cross-section when the steel is stretched [5]. Scholars have established the relationship between the nominal yield augmentation f_{vyv} of the corroded counterbearing and the rust loss rate η_{sv} of the counterbearing section as [6]:

$$f_{vyv} = \frac{a_1 - a_2 \eta_{sv}}{a_3 - \eta_{sv}} f_{vy} \quad (2)$$

where $\eta_{sv} = (A_v - A_{vc})/A_v$. A_v is the cross-sectional area of the counterbearing before rusting and A_{vc} is the cross-sectional area of the counterbearing after corrosion.

$$\rho_{sc} = A_{sc}/(bh_0); \rho_{vc} = A_{vc}/(bs) \quad (3)$$

where b is the section width of the beam and A_{sc} is the cross-sectional area of the corroded longitudinal bars [7]. Under the action of external load P , when the oblique section of the beam RC undergoes shear failure, the typical crack distribution is as shown in Figure 1 [8]. The specific image is shown in Figure 2. According to MCFT, we can establish stress balance conditions as:

$$f_1 = \begin{cases} E_c \varepsilon_1 & \varepsilon_1 \leq \varepsilon_{cr} \\ \frac{0.33\sqrt{f'_c}}{1+\sqrt{500}\varepsilon_1} & \varepsilon_1 > \varepsilon_{cr} \end{cases} \quad (4)$$

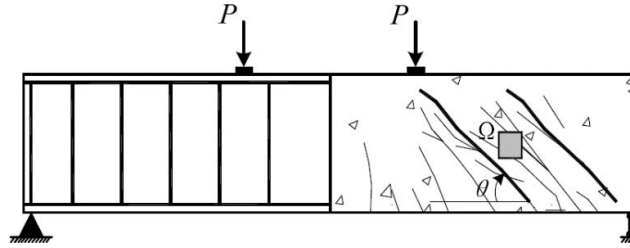


Fig. 1 RC beam with diagonal cracks. RC, reinforced concrete

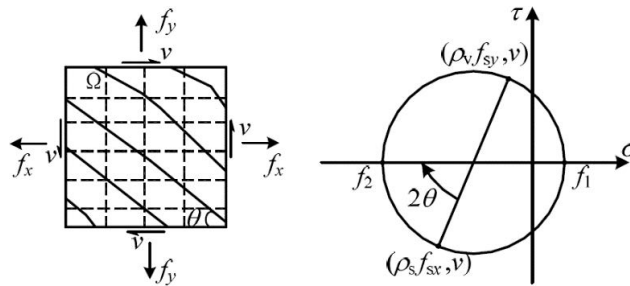


Fig. 2 The stress balance condition and stress Mohr circle of the tiny body

where ε_{cr} is the cracking strain of concrete, f'_c is the compressive augmentation of concrete and ε_1 is the primary tensile strain [9]. The force situation of the beam RC is shown in Figure 3, which can be obtained from the vertical force balance condition as:

$$A_v f_{vyc} (f_2 \sin^2 \theta - f_1 \cos^2 \theta) bs \quad (5)$$

The formula A_v is the cross-sectional area of the counterbearing.

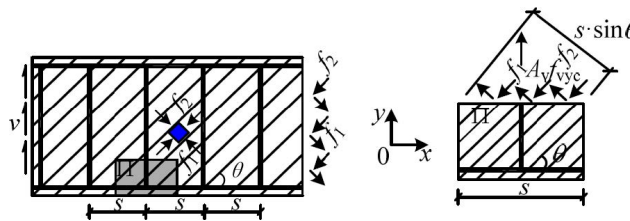


Fig. 3 Schematic diagram of beam principal stress and force balance

Combining formulas (4) and (5) we obtain as follows:

$$\begin{aligned} V = vbh_v &= \frac{f_1 bh_v}{\tan \theta + \cot \theta} + \frac{bh_v}{\tan \theta + \cot \theta} \left(\frac{A_v f_{vyc}}{bs} + f_1 \cos^2 \theta \right) \frac{1}{\sin^2 \theta} \\ &= \frac{f_1 bh_v}{\tan \theta + \cot \theta} (1 + \cot^2 \theta) + \frac{A_v f_{vyc} bh_v}{bs (\tan \theta + \cot \theta)} \times \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta} = f_1 bh_v \cot \theta + \frac{A_v f_{vyc} h_v}{s} \cot \theta \end{aligned} \quad (6)$$

According to formula (6), it can be seen that the shear capacity V of the beam RC is related to the diagonal tension stress f_1 . Based on Eq. (4), it can be seen that the diagonal tension stress f_1 is related to the principal tension strain ε_1 . ε_1 is related to the spacing of oblique cracks and the critical oblique crack dip angle θ . Therefore, the iterative calculation is needed to solve Eq. (6), and the calculation process is cumbersome as

$$\varepsilon_1 = a_7 \varepsilon_y = a_7 f_{vyc} / E_s \quad (7)$$

where a_7 is the influence coefficient of counterbearings and $\varepsilon_y = f_{vy}/E_s$ is the yield strain of the counterbearing. Corrosion of steel bars often causes concrete expansion, cracking or spalling, which reduces the effective shear cross-sectional area of RC beams [10]. The corrosion rate of the counterbearing section is as follows:

$$b_c = \begin{cases} b, & \eta_{sv} \leq 30\% \\ b - 2(c + d_{sv}) + \frac{s}{5.5}, & \eta_{sv} > 30\%, s \leq 5.5c \\ b - \frac{5.5}{s}(c + d_{sv})^2, & \eta_{sv} > 30\%, s > 5.5c \end{cases} \quad (8)$$

where d_{sv} is the diameter of the counterbearing. Combining formulas (1), (4), (7) and (8) the following is obtained:

$$V_n = V_c + V_s = f_1 b_c h_v \cot \theta + \frac{A_v f_{vy} h_v}{s} \cot \theta = \frac{0.33 b_c h_v \sqrt{f'_c}}{1 + \sqrt{500 a_7 \frac{f_{vy}}{E_s}}} \cot \theta + f_{vy} \frac{A_v}{s} h_v \cot \theta \quad (9)$$

Equation (9) cannot consider the objective uncertainties of the RC beam geometric dimensions, material properties, boundary constraints and other factors, which will lead to a certain degree of discreteness in the calculation results. Therefore, it is necessary to comprehensively consider the effects of objective uncertainty and subjective uncertainty.

3 A probabilistic model for calculating the shear capacity of corroded RC beams

3.1 Establishment of probabilistic shear capacity model

The values of the parameter a_1, a_2, a_3 in formula (2), parameter a_4, a_5, a_6 in formula (2), and parameter a_7 in formula (7) are mainly determined based on the engineering experience or test data fitting analysis. This paper introduces the probability model parameter $\beta_1 \sim \beta_7$ to consider the impact of objective uncertainty. The probability model parameter β_8 is introduced to consider the impact of subjective uncertainty as

$$V_{ncp} = \frac{0.33 b h_v \sqrt{f'_c}}{1 + \sqrt{500 \beta_7 \frac{f_{vy}}{E_s}}} \cot \theta_p + \frac{A_v f_{vy} h_v}{s} \cot \theta_p + \beta_8 + \xi \sigma \quad (10)$$

$$f_{vy} = \frac{\beta_1 - \beta_2 \eta_{sv}}{\beta_3 - \eta_{sv}} \quad (11)$$

$$\theta_p = (\beta_5 \lambda + \beta_6) \cdot \arctan \left(\sqrt{\frac{-\beta_4 k_{nsc} + \sqrt{a_4^2 k_{nsc}^2 + 4(1 - \beta_4) k_{nsc} k_{nvc}}}{2(1 - \beta_4) k_{nvc}}} \right) \quad (12)$$

where V_{ncp} is the probability value of the shear capacity of the corroded RC beam, $\xi \sigma$ is the systematic error of the probability model, ξ is a standard normal distribution random variable and σ is the standard deviation of the systematic error.

3.2 Determination of probability model parameters

According to the engineering experience or existing test data, the prior distribution of the probability model parameter β can be determined. Then the posterior distribution of β can be determined using Bayesian theory as:

$$P(\beta|V_{ncp}) = \frac{P(\beta)P(V_{ncp}|\beta)}{\int P(\beta)P(V_{ncp}|\beta)dV_{ncp}} \quad (13)$$

where $P(\beta|V_{ncp})$ is the posterior distribution probability density function of β , $P(\beta)$ is the primary distribution probability density function of β , $P(V_{ncp}|\beta)$ is the likelihood function of β and $\int P(\beta)P(V_{ncp}|\beta)dV_{ncp}$ is the normalisation factor. For this reason, this paper uses the Markov chain Monte Carlo (MCMC) method to determine the posterior distribution information of β . The basic parameters are shown in Table 1.

Table 1 Basic parameters of corroded RC beams and test values of shear capacity test

Numbering	b/mm	h/mm	λ	$\rho 1/(\%)$	fvy/Mpa	Vn1	Vn2	Vn3	Vn4
1	100	175	1.5	1.94	0.44	49.71	52.87	23.68	27.07
2	100	175	2.5	1.94	0.44	50.71	48.93	21.47	38
3	100	175	1.5	1.94	0.44	59.92	50.16	22.78	26.69
4	100	175	2.5	1.94	0.44	52.44	48.96	20.21	37.92
5	100	175	2.5	1.94	0.44	54.05	51.79	20.75	38.84
6	100	175	1.5	1.94	0.44	61.91	58.13	22.99	28.22
7	100	175	2.5	1.94	0.44	53.28	54.07	21.05	39.91
8	100	175	2.5	1.94	0.44	53.25	49.48	20.22	38.7
9	100	175	2.5	1.94	0.44	52.79	48.96	20.21	38.21
10	100	175	2.5	1.94	0.44	53.65	50.5	20.5	38.7
11	100	175	1.5	1.94	0.44	62.65	65.86	22.74	29.84
12	100	175	1.5	1.94	0.44	63.53	67.28	22.99	30.02
13	100	175	1.5	1.94	0.44	65.29	74.67	22.99	31.97
14	100	175	2.5	1.94	0.44	53.73	48.96	20.21	38.98
15	100	175	1.5	1.94	0.44	66.06	76.02	23.19	32.12
16	100	175	1.5	1.94	0.44	66.76	77.86	23.36	33.26
17	120	200	2	1.92	0.32	73.84	52.86	30.01	39.34
18	100	175	2.5	1.94	0.44	58.13	56.18	21.66	41.39
19	100	175	1.5	1.94	0.44	68.06	77.52	23.36	33.53
20	120	200	2	1.92	0.32	77.74	73.12	30.32	42.34

Based on the experimental data in Table 1 combined with formula (13) and the MCMC method, we can determine the posterior distribution information of each probability model parameter. The paper first uses $K-S$, a test to determine the empirical probability distribution type of the probability model parameter β . When the significance level is 0.05, and the sample size is 1,000, the critical value of the $K-S$ test is 0.043. The D value of $K-S$ test of each probability model parameter is shown in Table 2.

Table 2 Probability model parameter $K-S$ test D value

Parameter	β_1	β_2	β_3	β_4
Lognormal distribution	0.032	0.047	0.049	0.05
Weibull distribution	0.067	0.053	0.058	0.059
index distribution	0.055	0.506	0.541	0.56
Gamma distribution	0.031	0.039	0.045	0.048
Normal distribution	0.03	0.037	0.036	0.042
Parameter	β_5	β_6	β_7	β_8
Lognormal distribution	$\times$	0.038	0.064	$\times$
Weibull distribution	$\times$	0.068	0.048	$\times$
Index distribution	$\times$	0.586	0.473	$\times$
Gamma distribution	$\times$	0.037	0.053	$\times$
Normal distribution	0.031	0.035	0.034	0.025

The distribution type with the smallest D value is used as the posterior distribution of each probability model parameter. We found that none of the probability model parameters refused to obey the normal distribution. The average value of β_4 is 0.46, which indicates that the contribution of concrete shear capacity accounts for about 46% of the RC beam shear capacity. The mean value of β_5 is negative, which indicates that the critical oblique crack inclination angle of the RC beam shows a decreasing trend with the increase in the shear-span ratio.

4 Comparative analysis and verification

4.1 Validation of the probabilistic shear capacity model

Based on determining the posterior distribution information of the probability model parameter β , we can determine the probability statistical characteristic value of the shear bearing capacity of the RC beam and the confidence interval of different confidence levels by combining Eq. (13) with the MCS method. Taking the 50% and 95% confidence intervals as an example, the comparative analysis of the mean value (V_m) of the probability model and the test value (V_t) of the experiment is shown in Figure 4.

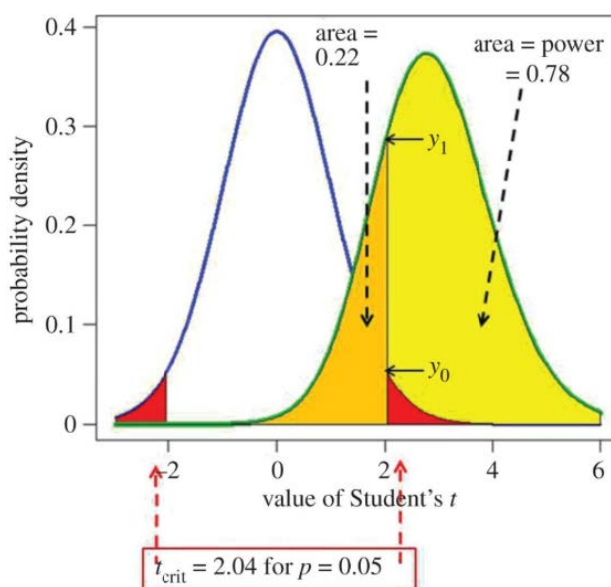


Fig. 4 The distribution of experimental test values in the confidence interval of the probability model

It can be seen from Figure 4 that nearly 1/2 of the measured values are within the 50% confidence interval, and almost all of the measured values are within the 95% confidence interval. This shows that the established probability model can better describe the probability distribution characteristics of the beam's shear capacity.

4.2 Calibration analysis of deterministic shear capacity model

Taking the deterministic shear capacity calculation model as an example, the distribution of the calculated values of each model and the experimental test values within the confidence interval of the probability model is as shown in Figure 5.

Although nearly half of the calculated value of the model V_{n1} falls within the 95% confidence interval, the overall deviation is far from the measured value. This shows that the dispersion is large, and the calculation accuracy is limited. The calculation results of the V_{n2} model are highly discrete, and most of the data points fall outside the 95% confidence interval. This shows that the calculation accuracy of the model is low.

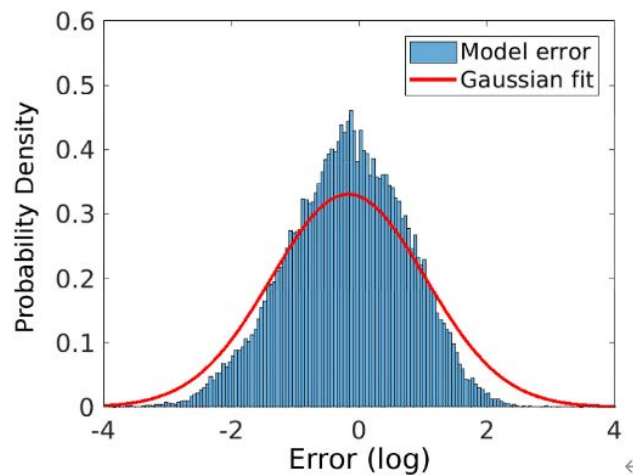


Fig. 5 The distribution of the calculated value of the deterministic model in the confidence interval of the probability model

In addition, we can use the probability model of this paper to calibrate the calculation accuracy of the traditional deterministic model. Choose a corroded beam from Table 1 as an example. The mean value and standard deviation of the shear capacity determined by the probability model are 52.64 kN and 8.77 kN, respectively. After the K–S test, the shear bearing capacity does not refuse to obey the normal distribution, and its probability density distribution is as shown in Figure 6. The calculated values of models V_{n1} , V_{n2} , V_{n3} , and V_{n4} are 58.13 kN, 56.18 kN, 21.66 kN, and 41.39 kN, respectively. The calculated values of the V_{n1} and V_{n2} models are both more significant than the average, and the calculated values of the V_{n3} and V_{n4} models are both less than the average. The calculated value of the V_{n3} model is too small. It can be seen that the probability model established in this paper can be used to calibrate the calculation accuracy of the traditional deterministic shear capacity model.

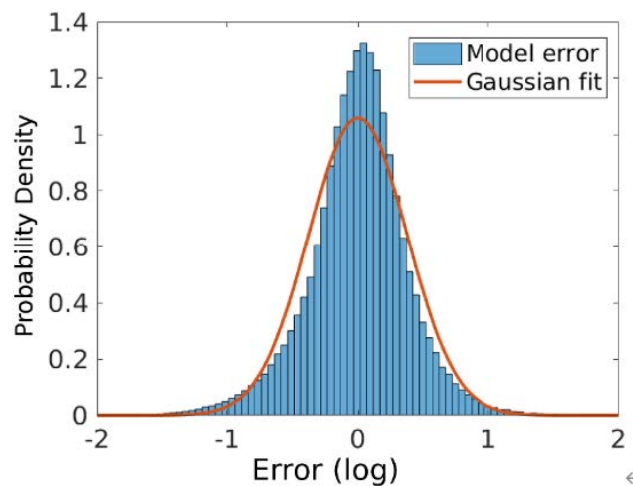


Fig. 6 Calibration of the calculation accuracy of the deterministic model

5 Conclusion

This paper combined the modified pressure field theory, Bayesian theory and MCMC method to establish a probability model for calculating the shear capacity of corroded beams. The established probability model for

calculating the shear capacity of corroded beams not only has a rigorous theoretical basis but also comprehensively considers the effects of subjective and objective uncertainties. It has good applicability and calculation accuracy, which can reasonably describe the probability distribution characteristics of the shear capacity of corroded beams.

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